THOUGHTSPOT

IIT Guwahati | Programming Problem Statements

Problem 1: Coupled Recurrence Relations

Problem Statement

You are given integers **a, b, c, d, e, f,** and **n**. Three sequences **F**, **G**, and **H** are defined by the following recurrence relations:

$$F(n) = a \times F(n-1) + b \times G(n-2)$$

$$G(n) = c \times G(n-1) + d \times H(n-2)$$

$$H(n) = e \times H(n-1) + f \times F(n-2)$$

The initial values required to calculate the three sequences are also given. Find the value of **F(n)** modulo $10^9 + 7$.

Input

- Seven integers: a, b, c, d, e, f, and n.
- The initial values required to calculate the three sequences.

Output

Print F(n) modulo 1,000,000,007.

Constraints

- $0 \le n \le 10^9$.
- The constraints on the coefficients and initial values are as specified in the original problem.

Problem 2: Tray Candy Game

Problem Statement

There are **n** trays arranged in a stack, one on top of another. The **i**-th tray contains **a[i]** candies.

Two players, A and S, play a game. Player A makes the first move. Before the game begins, player S can rearrange the n trays in any order.

Game Rules

1. On each turn, the current player must take at least one candy from the top tray.
2. A player can take any positive number of candies, up to all the candies remaining in the top tray.
3. If a tray becomes empty, it is removed from the stack, and the next tray becomes accessible.
4. Players take turns alternately.
5. The player who cannot take any candies on their turn loses.
6. Both players play optimally.

Task

Count the number of permutations of the n trays for which player A wins. Since the answer may be large, print it modulo 1,000,000,007.

Input

- An integer n, representing the number of trays.
- An array a[1], a[2], …, a[n], where a[i] represents the number of candies in the i-th tray.

Output

Print the number of permutations of the trays for which player A wins, modulo 1,000,000,007.